

Data Definition

Engagement data

The engagement data are aggregated at school district level, and each file in the folder engagement_data represents data from one school district. The 4-digit file name represents district_id which can be used to link to district information in district_info.csv. The lp_id can be used to link to product information in product_info.csv.

Name	Description
time	date in "YYYY-MM-DD"
lp_id	The unique identifier of the product
pct_access	Percentage of students in the district have at least one page-load event of a given product and on a given day
engagement_index	Total page-load events per one thousand students of a given product and on a given day

District information data

The district file districts_info.csv includes information about the characteristics of school districts, including data from [NCES](#) (2018-19), [FCC](#) (Dec 2018), and [Edunomics Lab](#). In this data set, we removed the identifiable information about the school districts. We also used an open source tool [ARX \(Prasser et al. 2020\)](#) to transform several data fields and reduce the risks of re-identification. For data generalization purposes some data points are released with a range where the actual value falls under. Additionally, there are many missing data marked as 'NaN' indicating that the data was suppressed to maximize anonymization of the dataset.

Name	Description
district_id	The unique identifier of the school district
state	The state where the district resides in
locale	NCES locale classification that categorizes U.S. territory into four types of areas: City, Suburban, Town, and Rural. See Locale Boundaries User's Manual for more information.
pct_black/hispanic	Percentage of students in the districts identified as Black or Hispanic based on 2018-19 NCES data
pct_free/reduced	Percentage of students in the districts eligible for free or reduced-price lunch based on 2018-19 NCES data
county_connections_ratio	ratio (residential fixed high-speed connections over 200 kbps in at least one direction/households) based on the county level data from FCC Form 477 (December 2018 version). See FCC data for more information.
pp_total_raw	Per-pupil total expenditure (sum of local and federal expenditure) from Edunomics Lab's National Education Resource Database on Schools (NERD\$) project. The expenditure data are school-by-school, and we use the median

Name	Description
	value to represent the expenditure of a given school district.

Product information data

The product file products_info.csv includes information about the characteristics of the top 372 products with most users in 2020. The categories listed in this file are part of LearnPlatform's product taxonomy. Data were labeled by our team. Some products may not have labels due to being duplicate, lack of accurate url or other reasons.

Name	Description
LP ID	The unique identifier of the product
URL	Web Link to the specific product
Product Name	Name of the specific product
Provider/Company Name	Name of the product provider
Sector(s)	Sector of education where the product is used

Name	Description
Primary Essential Function	The basic function of the product. There are two layers of labels here. Products are first labeled as one of these three categories: LC = Learning & Curriculum, CM = Classroom Management, and SDO = School & District Operations. Each of these categories have multiple sub-categories with which the products were labeled

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